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Next item →

1.

What is the purpose of the interacting multiple model (IMM) Kalman filter?

1 / 1 point

☒ To track an object that may operate in different modes over different periods of time.

☐ To track an object that obeys an NCV model.

☐ To track an object that obeys an NCP model.

☐ To track an object that obeys a CT model.

☒ Correct

Yes. This is correct.

2.

In the IMM Kalman filter, what is the interpretation of the  $\mu_{j,k}$  vector?

1 / 1 point

☐ It is the conditional probability of arriving to mode  $j$  at time  $k$ .

☐ It is the likelihood of the present measurement given that the target is operating in mode  $j$ .

☐ It is the overall state estimate at time  $k$ .

☒ It is the pmf describing the probability that the target is operating in each mode at time  $k$ .

☒ Correct

Yes. That is correct.

3.

In the IMM Kalman filter, what is the interpretation of the  $\Lambda_{j,k}$  vector?

1 / 1 point

☐ It is the conditional probability of arriving to mode  $j$  at time  $k$ .

☒ It is the likelihood of the present measurement given that the target is operating in mode  $j$ .

☐ It is the overall state estimate at time  $k$ .

☐ It is the pmf describing the probability that the target is operating in each mode at time  $k$ .

☒ Correct

Yes. That is correct.